

The global-closure obstruction (G) is a cohomology class:
the Zappa–Szép holonomy as $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$

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Abstract

We resolve the next layer of SEED-Q2. The two-level criterion for a strict factorization $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ over a wide subcategory $\mathcal{D}$ splits into a fibrewise freeness condition (L) and a global closure condition (G); the prior notes proved $(L) \wedge (G)$ decides the question but described (G) only as “an obstruction.” Here we *classify* it. Assuming (L), every transversal (choice of free bases) carries a *defect 2-cochain* ω valued in $\mathcal{D}$, with (G) holding iff $\omega \equiv \mathrm{id}$. We isolate a clean hypothesis (H) — abelian vertex groups and a *trivial left action* — under which the free $\mathcal{D}$ -orbits form a category $\mathrm{Sk}_{\mathcal{C}}$, the vertex automorphism groups form a presheaf $\mathcal{D}: \mathrm{Sk}_{\mathcal{C}}^{\mathrm{op}} \rightarrow \mathbf{Ab}$, the defect is a normalized 2-cocycle, and re-choosing the transversal changes it by a coboundary. Hence $[\omega] \in H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ is a transversal-independent invariant and

$$(G) \text{ holds} \iff [\omega] = 0.$$

The set of strict factorizations, when nonempty, is a torsor under $Z^1(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$, and modulo inner relabelings under $H^1(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D})$ — the H^1/H^2 pattern the conjecture predicted. We compute the ten-morphism rigid twist explicitly: $\mathrm{Sk}_{\mathcal{C}}$ is the branch $a \rightarrow x \rightrightarrows y$, $\mathcal{D}$ is $\mathbb{Z}/2$ concentrated at a , $H^2(\mathrm{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$, and the defect represents the *nonzero* element; the cross-arrow enlargement of the pairwise counterexample gives the same class. Finally we delimit honestly: a nontrivial left action (which already occurs in two-morphism examples) breaks the descent to $\mathrm{Sk}_{\mathcal{C}}$, and the obstruction’s natural home becomes the *nonabelian* H^2 of the matched pair — the Kac/Rosebrugh–Wood regime, which we state as the open continuation. All claims are machine-checked (`projects/scratch/cohomology_holonomy.py`).

1 Setup

Throughout, $\mathcal{K}$ is a small category and $\mathcal{D} \subseteq \mathcal{K}$ a wide subcategory (the prescribed right factor). We use the pairwise criterion of the companion note [1] freely: $\mathcal{K} = \mathcal{C} \bowtie \mathcal{D}$ for some wide $\mathcal{C}$ iff

(L) each hom-presheaf $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is free as a right $\mathcal{D}$ -set; and

(G) free bases $B_b \subseteq \mathrm{Hom}_{\mathcal{K}}(-, b)$ can be chosen so that $\mathcal{C} = \bigcup_b B_b$ is a wide subcategory.

We *assume* (L) *throughout*. Our object of study is the residual condition (G).

Recall the structure of a free $\mathcal{D}$ -set. The right $\mathcal{D}$ -action on $\mathrm{Hom}_{\mathcal{K}}(-, b)$ is precomposition, $f \cdot d = f \circ d$ for $d \in \mathcal{D}$. An *orbit* is a connected component of the action; (L) says each orbit is free. A morphism $c: a \rightarrow b$ *generates* its orbit o iff $d \mapsto c \circ d$ is a bijection from the $\mathcal{D}$ -arrows into a onto o ; equivalently every f in o factors uniquely as $f = c \circ d$, $d \in \mathcal{D}$. We call $a = \mathrm{dom}(c)$ the *apex* of o (well defined up to $\mathcal{D}$ -isomorphism). Two generators of o differ by a $\mathcal{D}$ -isomorphism on the right: $c' = c \circ \varphi$ with φ a $\mathcal{D}$ -iso into the apex. We write $\mathrm{orb}(f)$ for the orbit of f .

Definition 1 (Vertex groups). For an object a put $G_a := \text{Aut}_{\mathcal{D}}(a)$, the group of $\mathcal{D}$ -automorphisms of a . When $\mathcal{D}$ is a disjoint union of vertex groups (the single-category case $\mathcal{D} = \mathcal{E}$ with invertible endomorphisms), $G_a = \text{End}_{\mathcal{D}}(a)$ and apex = source.

Definition 2 (Transversal). A *transversal* T is a choice of one generator $c_o \in o$ for every orbit o (over all targets b), subject to the *unit normalization*: at the orbit of id_b (which is the regular orbit $\mathcal{D}(-, b)$, generated by the $\mathcal{D}$ -automorphisms of b) we choose $c = \text{id}_b$. The associated candidate left factor is $\mathcal{C}_T := \{c_o\}_o$, the set of chosen generators.

By [1, Thm. 1], $\mathcal{C}_T$ is the left factor of a strict factorization system $(\mathcal{C}_T, \mathcal{D})$ iff $\mathcal{C}_T$ is closed under composition; this is exactly (G) for the choice T .

2 The defect 2-cochain (general)

The following is unconditional (it needs only (L) and a transversal).

Definition 3 (Defect). Fix a transversal T . For composable generators $c_1: a \rightarrow b$ and $c_2: b \rightarrow e$ in $\mathcal{C}_T$, the composite $c_2 \circ c_1: a \rightarrow e$ lies in some orbit at e ; let $\lfloor c_2 c_1 \rfloor_T \in T$ be that orbit's chosen generator. By freeness there is a *unique* $\omega_T(c_2, c_1) \in \mathcal{D}$ with

$$c_2 \circ c_1 = \lfloor c_2 c_1 \rfloor_T \circ \omega_T(c_2, c_1), \quad \omega_T(c_2, c_1): a \rightarrow \text{dom} \lfloor c_2 c_1 \rfloor_T. \quad (1)$$

We call ω_T the *defect 2-cochain* of T .

Proposition 4 (Closure = vanishing of the defect). (G) holds for the transversal T (i.e. $\mathcal{C}_T$ is a wide subcategory) if and only if $\omega_T(c_2, c_1)$ is an identity for every composable pair.

Proof. $\mathcal{C}_T$ contains all identities (unit normalization) and is closed under composition iff $c_2 \circ c_1 \in \mathcal{C}_T$ for all composable generators. By (1), $c_2 \circ c_1 = \lfloor c_2 c_1 \rfloor_T \circ \omega_T(c_2, c_1)$ with $\lfloor c_2 c_1 \rfloor_T \in \mathcal{C}_T$; since a generator together with a nonidentity $\mathcal{D}$ -arrow gives, by uniqueness of factorization, an element *outside* the generating set unless the $\mathcal{D}$ -arrow is an identity, we have $c_2 \circ c_1 \in \mathcal{C}_T \iff \omega_T(c_2, c_1) = \text{id}$. (If $\omega_T(c_2, c_1) = \text{id}$ then $c_2 c_1 = \lfloor c_2 c_1 \rfloor_T \in \mathcal{C}_T$; conversely if $c_2 c_1 \in \mathcal{C}_T$ it is a chosen generator, hence *the* generator $\lfloor c_2 c_1 \rfloor_T$ of its orbit, forcing $\omega_T(c_2, c_1) = \text{id}$ by uniqueness.) $\square$

Definition 5 (Gauge). The *gauge group* is $\mathcal{G} := \prod_o \Gamma_o$, where Γ_o is the group of $\mathcal{D}$ -automorphisms of the apex of o acting on generators of o by right multiplication $c_o \mapsto c_o \circ h_o$, and h is normalized ($h_o = \text{id}$ on unit orbits). $\mathcal{G}$ acts *freely and transitively* on the set of transversals: any two transversals differ by a unique gauge (replace each chosen generator by the other orbit's generator, which differ by a $\mathcal{D}$ -iso). Write $T \cdot h$ for the regauged transversal.

3 The clean hypothesis and the cochain complex

The gauge transformation of ω involves *post*composition by target automorphisms — the left action of the matched pair — and is nonabelian in general. We isolate the regime in which it is a coboundary in an abelian complex.

Hypothesis (H).

- (i) (*abelian vertex groups*) $\mathcal{D}$ is a disjoint union of vertex groups, $\mathcal{D}(a, a) = G_a = \text{End}_{\mathcal{D}}(a)$ a group and $\mathcal{D}(a, b) = \emptyset$ for $a \neq b$, with every G_a abelian; equivalently $\mathcal{K}$ is a single small category with invertible endomorphisms and $\mathcal{D} = \mathcal{E}$ [2];

- (ii) (*trivial left action*) for every generator $c: a \rightarrow b$ and every $\psi \in G_b$, the morphism $\psi \circ c$ lies in $\text{orb}(c)$ — equivalently $\psi \circ c = c \circ \varphi_c(\psi)$ for a (necessarily unique) $\varphi_c(\psi) \in G_a$.

Under (H)(i) the apex of every orbit equals the common source of its elements (the $\mathcal{D}$ -arrows into a are exactly G_a , all $a \rightarrow a$), and for composable generators $c_1: a \rightarrow b$, $c_2: b \rightarrow e$ the composite $c_2 \circ c_1$ *generates* its orbit: the orbit is $\{c_2 c_1 \circ \gamma : \gamma \in G_a\}$ and $\gamma \mapsto c_2 c_1 \circ \gamma$ is a bijection by freeness (L). Hence the defect $\omega_T(c_2, c_1)$ of (1) is forced to lie in the *group* $G_a = G_{\text{dom } c_1}$ — the coefficient we cohomologize.

Lemma 6 (Descent to the orbit category). *Under (H)(ii), the assignment $o_2 * o_1 := \text{orb}(c_2 \circ c_1)$ (any generators c_i of o_i , composable) is well defined and associative, and makes $\text{Sk}_{\mathcal{C}}$ — objects $\text{ob } \mathcal{K}$, morphisms the free $\mathcal{D}$ -orbits, identities the unit orbits — a category. Under (H) the rule $a \mapsto G_a$ on objects and $\varphi_c: G_b \rightarrow G_a$ on a generator $c: a \rightarrow b$ extends to a presheaf $\mathcal{D}: \text{Sk}_{\mathcal{C}}^{\text{op}} \rightarrow \mathbf{Ab}$ (a functor on $\text{Sk}_{\mathcal{C}}^{\text{op}}$).*

Proof. Well-defined composition. Replacing c_1 by $c_1 \circ \varphi$ (φ a $\mathcal{D}$ -iso into the apex) leaves $\text{orb}(c_2 \circ c_1 \circ \varphi) = \text{orb}(c_2 \circ c_1)$ (right $\mathcal{D}$ -translation preserves orbits). Replacing c_2 by $c_2 \circ \psi$ ($\psi \in G_b$, b the apex of o_2): by (H)(ii), $\psi \circ c_1 = c_1 \circ \varphi_{c_1}(\psi)$, so $c_2 \circ \psi \circ c_1 = c_2 \circ c_1 \circ \varphi_{c_1}(\psi)$, again the same orbit. Associativity and unitality of $*$ are inherited from $\mathcal{K}$ because $\text{orb}(-)$ depends only on the underlying $\mathcal{K}$ -morphism, which associates.

Presheaf. $\varphi_c: G_b \rightarrow G_a$ is a homomorphism: for $\psi, \psi' \in G_b$, $c \circ \varphi_c(\psi' \psi) = (\psi' \psi) \circ c = \psi' \circ (\psi \circ c) = \psi' \circ c \circ \varphi_c(\psi) = c \circ \varphi_c(\psi') \varphi_c(\psi)$, and cancel c on the left (it is a generator, hence left-cancellable against distinct G_a -translates). Functoriality $\varphi_{c_2 * c_1} = \varphi_{c_1} \circ \varphi_{c_2}$ is the same calculation applied to $c_2 \circ c_1$, using (H)(ii) twice; $\varphi_{\text{id}} = \text{id}$. Contravariance is built into the typing $G_b \rightarrow G_a$ along $c: a \rightarrow b$. $\square$

We use the standard cochain complex of a small category with coefficients in a presheaf [4, 5]. For $n \geq 0$,

$$C^n(\text{Sk}_{\mathcal{C}}; \mathcal{D}) = \prod_{o_n, \dots, o_1 \text{ composable}} G_{\text{dom } o_1},$$

$$(\delta\sigma)(o_{n+1}, \dots, o_1) = \varphi_{o_1}(\sigma(o_{n+1}, \dots, o_2)) + \sum_{i=1}^n (-1)^i \sigma(o_{n+1}, \dots, o_{i+1} * o_i, \dots, o_1) + (-1)^{n+1} \sigma(o_n, \dots, o_1),$$

the first face dropping the rightmost arrow o_1 and acting by its restriction, the last dropping the leftmost o_{n+1} ; normalized cochains vanish whenever some o_i is a unit. Concretely

$$(\delta^1 h)(o_2, o_1) = \varphi_{o_1}(h(o_2)) - h(o_2 * o_1) + h(o_1), \quad (2)$$

$$(\delta^2 \omega)(o_3, o_2, o_1) = \varphi_{o_1}(\omega(o_3, o_2)) - \omega(o_3, o_2 * o_1) + \omega(o_3 * o_2, o_1) - \omega(o_2, o_1). \quad (3)$$

$\delta^2 \circ \delta^1 = 0$ (verified directly: the eight terms cancel using $\varphi_{o_2 * o_1} = \varphi_{o_1} \circ \varphi_{o_2}$); $H^n(\text{Sk}_{\mathcal{C}}; \mathcal{D}) := \ker \delta^n / \text{im } \delta^{n-1}$.

4 Main theorem

A transversal is the same thing as a section of $\text{Sk}_{\mathcal{C}}$ over $\mathcal{K}$ (a choice of generator per orbit); the gauge group of Definition 5 is exactly the group $C^1(\text{Sk}_{\mathcal{C}}; \mathcal{D})$ of normalized 1-cochains, $h: o \mapsto h_o \in G_{\text{dom } o}$.

Theorem 7 (The closure obstruction is a cohomology class). *Assume (L) and (H). Then:*

1. the defect ω_T descends to a normalized 2-cochain on $\text{Sk}_{\mathcal{C}}$ and is a 2-cocycle: $\delta^2 \omega_T = 0$;

2. regauging changes it by a coboundary: $\omega_{T \cdot h} = \omega_T + \delta^1 h$ for every $h \in C^1(\text{Sk}_{\mathcal{C}}; \mathcal{D})$;

3. consequently $[\omega] := [\omega_T] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$ is independent of the transversal T , and

$$(G) \text{ holds} \iff [\omega] = 0;$$

4. if nonempty, the set of strict factorization systems $(\mathcal{C}, \mathcal{D})$ of $\mathcal{K}$ is a torsor under $Z^1(\text{Sk}_{\mathcal{C}}; \mathcal{D})$; modulo the inner relabelings $B^0 = \text{im } \delta^0$ (automorphisms of $(\mathcal{K}, \mathcal{D})$ fixing objects) it is a torsor under $H^1(\text{Sk}_{\mathcal{C}}; \mathcal{D})$.

Proof. Throughout, additive notation in the abelian groups G_a . By the remark after (H), the defect $\omega_T(c_2, c_1)$ lands in the group $G_{\text{dom } c_1}$. It depends only on the orbits $o_i = \text{orb}(c_i)$: changing $c_1 \mapsto c_1 \gamma$ ($\gamma \in G_a$) multiplies both sides of (1) on the right by γ and rescales the defect accordingly; changing $c_2 \mapsto c_2 \psi$ ($\psi \in G_b$) uses (H)(ii), $\psi \circ c_1 = c_1 \circ \varphi_{c_1}(\psi)$, to absorb the change into the right G_a -action. So ω_T is a normalized 2-cochain on $\text{Sk}_{\mathcal{C}}$.

(1) *Cocycle.* Factor $c_3 \circ c_2 \circ c_1$ two ways and use uniqueness. Writing $[\cdot]$ for $[\cdot]_T$ and ω for ω_T :

$$\begin{aligned} c_3 \circ (c_2 \circ c_1) &= c_3 \circ ([c_2 c_1] \omega(c_2, c_1)) = (c_3 \circ [c_2 c_1]) \omega(c_2, c_1) \\ &= [c_3 [c_2 c_1]] \omega(c_3, [c_2 c_1]) \omega(c_2, c_1), \end{aligned}$$

while

$$\begin{aligned} (c_3 \circ c_2) \circ c_1 &= ([c_3 c_2] \omega(c_3, c_2)) \circ c_1 = [c_3 c_2] \circ (\omega(c_3, c_2) \circ c_1) \\ &= [c_3 c_2] \circ c_1 \circ \varphi_{c_1}(\omega(c_3, c_2)) = [[c_3 c_2] c_1] \omega([c_3 c_2], c_1) \varphi_{c_1}(\omega(c_3, c_2)), \end{aligned}$$

the third equality by (H)(ii). The two leading generators agree ($[c_3 [c_2 c_1]] = [c_3 c_2 c_1] = [[c_3 c_2] c_1]$ since $*$ is associative in $\text{Sk}_{\mathcal{C}}$). By uniqueness of the $\mathcal{D}$ -part,

$$\omega(c_3, [c_2 c_1]) + \omega(c_2, c_1) = \omega([c_3 c_2], c_1) + \varphi_{c_1}(\omega(c_3, c_2)),$$

which is exactly $(\delta^2 \omega)(o_3, o_2, o_1) = 0$ in (3).

(2) *Coboundary.* Regauge $c_i \mapsto c_i h_{o_i}$, $h_{o_i} \in G_{\text{dom } c_i}$. Then $[\cdot] \mapsto [\cdot] h_{o_2 * o_1}$ and

$$\begin{aligned} (c_2 h_{o_2}) \circ (c_1 h_{o_1}) &= c_2 \circ (h_{o_2} \circ c_1) \circ h_{o_1} = c_2 \circ c_1 \circ \varphi_{c_1}(h_{o_2}) h_{o_1} \\ &= [c_2 c_1] \omega_T(c_2, c_1) \varphi_{o_1}(h_{o_2}) h_{o_1} = ([c_2 c_1] h_{o_2 * o_1}) (h_{o_2 * o_1}^{-1} \omega_T(c_2, c_1) \varphi_{o_1}(h_{o_2}) h_{o_1}), \end{aligned}$$

using (H)(ii) for $h_{o_2} \circ c_1$. Reading off the $\mathcal{D}$ -part for the regauged transversal,

$$\omega_{T \cdot h}(o_2, o_1) = \omega_T(o_2, o_1) - h_{o_2 * o_1} + \varphi_{o_1}(h_{o_2}) + h_{o_1} = \omega_T(o_2, o_1) + (\delta^1 h)(o_2, o_1),$$

by (2).

(3) *Invariance and the equivalence.* Since the gauge acts transitively on transversals (Definition 5), (2) shows any two defects are cohomologous; with (1) this gives a well-defined class $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$. If (G) holds then some T has $\omega_T \equiv 0$ (Proposition 4), so $[\omega] = 0$. Conversely if $[\omega] = 0$ then $\omega_T = \delta^1 h$ for some h and any T ; by (2), $\omega_{T \cdot (-h)} = \omega_T - \delta^1 h = 0$, so $T \cdot (-h)$ closes and (G) holds.

(4) *Torsor.* If T closes ($\omega_T = 0$), then $T \cdot h$ closes iff $\omega_{T \cdot h} = \delta^1 h = 0$ iff $h \in Z^1$. Thus the closing transversals – equivalently the strict factorization systems $(\mathcal{C}_T, \mathcal{D})$ – form a torsor under $Z^1(\text{Sk}_{\mathcal{C}}; \mathcal{D})$. The 0-cochains $C^0 = \prod_a G_a$ act by δ^0 as the automorphisms of $(\mathcal{K}, \mathcal{D})$ that fix every object and rescale the $\mathcal{D}$ -action basepoints; quotienting by $B^0 = \text{im } \delta^0$ identifies SFS related by such an inner relabeling, leaving a torsor under $Z^1/B^0 = H^1(\text{Sk}_{\mathcal{C}}; \mathcal{D})$. $\square$

Remark 8 (Why this is the right complex). $\text{Sk}_{\mathcal{C}}$ is the would-be left factor $\mathcal{C}$ *regarded only up to the right $\mathcal{D}$ -action*: its morphisms are the orbits, i.e. the $\mathcal{C}$ -arrows “modulo the ambiguity of choosing a representative.” The presheaf $\mathcal{D}$ records the residual torsor of choices. Theorem 7 says precisely that the local choices (free, by (L)) glue to a global section (a closed $\mathcal{C}$) iff a single 2-dimensional obstruction vanishes — the categorical avatar of “a principal bundle with flat structure is trivial iff its holonomy class vanishes.” The companion equivalence “distributive law $\lambda: \mathcal{D}\mathcal{C} \Rightarrow \mathcal{C}\mathcal{D} \leftrightarrow \text{SFS}$ ” of Rosebrugh–Wood [3] is, in this language, the statement that the matched-pair cocycle conditions ZS1–ZS4 [1] are the simplicial identities of which (3) is the abelianized shadow.

5 The rigid twist computes the generator of $\mathbb{Z}/2$

Example 9 (Ten-morphism rigid twist). Objects a, x, y ; $\text{End}(a) = \{\text{id}_a, g\}$ with $g^2 = \text{id}_a$; $\text{End}(x), \text{End}(y)$ trivial; $\text{Hom}(a, x) = \{p, pg\}$, $\text{Hom}(a, y) = \{q, qg\}$, $\text{Hom}(x, y) = \{s, s_2\}$, with the rigid twist $s \circ p = q$, $s_2 \circ p = qg$ (forcing $s \circ pg = qg$, $s_2 \circ pg = q$). Take $\mathcal{D} = \mathcal{E}$ the endomorphisms, so $G_a = \mathbb{Z}/2$ and $G_x = G_y = 0$.

Theorem 10. *In Example 9, (L) and (H) hold, $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$, and the defect represents the nonzero element; hence (G) fails and $\mathcal{K}$ has no strict factorization over $\mathcal{E}$.*

Proof. (L) is the freeness of all six hom-sets over their source vertex group (verified in [2]). (H)(i) holds ($\mathbb{Z}/2$ abelian). (H)(ii): the only nontrivial G_b is G_a , and the only generator into a is id_a (the regular orbit), on which $g \circ \text{id}_a = \text{id}_a \circ g \in \text{orb}(\text{id}_a)$; all other targets have trivial G_b . So the left action is trivial.

The orbit category $\text{Sk}_{\mathcal{C}}$ has, besides identities, the morphisms $[p]: a \rightarrow x$, $[q]: a \rightarrow y$, and $[s], [s_2]: x \rightarrow y$, with $[s] * [p] = \text{orb}(s \circ p) = \text{orb}(q) = [q] = \text{orb}(s_2 \circ p) = [s_2] * [p]$. The presheaf is $\mathcal{D}(a) = \mathbb{Z}/2$, $\mathcal{D}(x) = \mathcal{D}(y) = 0$, all restriction maps zero (no nonidentity generator targets a).

Normalized cochains: C^1 has the two coordinates $h([p]), h([q]) \in \mathbb{Z}/2$ (generators with source-group $\mathbb{Z}/2$); C^2 has the two coordinates $\omega([s], [p]), \omega([s_2], [p]) \in \mathbb{Z}/2$ (composable pairs with $\text{dom} = a$); $C^3 = 0$ (y is a sink, so no composable triple starts at a). Hence $Z^2 = C^2 \cong (\mathbb{Z}/2)^2$. From (2) (all $\varphi = 0$),

$$(\delta^1 h)([s], [p]) = h([p]) - h([s] * [p]) = h([p]) - h([q]), \quad (\delta^1 h)([s_2], [p]) = h([p]) - h([q]),$$

so $B^2 = \{(t, t) : t \in \mathbb{Z}/2\}$, the diagonal. Therefore

$$H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) = (\mathbb{Z}/2)^2 / \langle (1, 1) \rangle \cong \mathbb{Z}/2.$$

For the transversal $T = \{p, q, s, s_2, \dots\}$ the defect is $\omega([s], [p]) = 0$ (as $s \circ p = q = \sqcup \circ \text{id}$) and $\omega([s_2], [p]) = g \mapsto 1$ (as $s_2 \circ p = qg = q \circ g$), i.e. $\omega_T = (0, 1) \notin B^2$. Thus $[\omega] \neq 0$ is the generator, and by Theorem 7(3), (G) fails. $\square$

The next proposition is *outside* the hypothesis of Theorem 7 (its $\mathcal{D}$ is not a disjoint union of vertex groups), yet exhibits the identical class — evidence that the vertex-group restriction in (H)(i) is an artifact of the proof, not of the phenomenon, and can likely be relaxed to “the defect is group-valued and the left action is trivial.”

Proposition 11 (The cross-arrow enlargement keeps the same class). *Adjoin an object w and a free $\langle g \rangle$ -torsor $\{d, gd\}: w \rightarrow a$ to $\mathcal{D}$, with the induced rows $\{pd, pgd\}: w \rightarrow x$, $\{qd, qgd\}: w \rightarrow y$ in $\mathcal{K}$ (the pairwise counterexample of [1]). Then $\mathcal{D}$ is not a groupoid (d is noninvertible), so (H)(i) fails; nonetheless (L) holds, the cross arrows are absorbed into existing orbits ($\text{orb}(p) = \{p, pg, pd, pgd\}$*

with apex a ; the only $\mathcal{C}$ -arrow into a is still id_a), the defect remains valued in $G_a = \mathbb{Z}/2$, the left action is trivial, and the construction of Theorem 7 goes through verbatim to give $H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D}) \cong \mathbb{Z}/2$ with the defect the nonzero class.

Proof. Direct computation (`cohomology_holonomy.py`): the $\mathcal{D}$ -orbit of p contains the w -row because $pd = p \circ d$ with $d \in \mathcal{D}$, so $\text{Sk}_{\mathcal{C}}$ is unchanged on a, x, y and adds only the source object w (a non-target of any nonidentity generator). Every defect lands in $\mathcal{D}(a, a) = G_a = \mathbb{Z}/2$ because each composable $\mathcal{C}$ -pair has the form (σ, c_1) with $\text{cod } \sigma = y$, $\text{dom } c_1 = a$ and $\sigma \circ c_1 \in \text{orb}(q)$ (apex a); the left action is trivial since the only nontrivial vertex group sits at the *source* a , never at a middle object. The $H^2 \cong \mathbb{Z}/2$ computation is then identical to Theorem 10. $\square$

6 Honest scope: the nonabelian boundary

Hypothesis (H)(ii) is not a technicality; it is the exact line past which the obstruction stops being an abelian H^2 .

Example 12 (Nontrivial left action in two morphisms). Objects a, b ; $G_a = 0$, $G_b = \{\text{id}_b, h\}$ with $h^2 = \text{id}_b$; $\text{Hom}(a, b) = \{t, ht\}$ with $ht := h \circ t$. Then $\text{Hom}(a, b)$ has two singleton free orbits $\{t\}$, $\{ht\}$, and the target automorphism h swaps them: $h \circ t = ht \notin \text{orb}(t)$. (H)(ii) fails.

When (H)(ii) fails, $\text{orb}(c_2 \circ c_1)$ depends on the chosen generator of o_2 , so $\text{Sk}_{\mathcal{C}}$ is not a category, the presheaf $\mathcal{D}$ does not exist, and the coboundary computation in Theorem 7(2) acquires a genuinely *nonabelian* correction: the regauging of ω mixes the left action $h_{o_2} \circ c_1$ with the right action, exactly as in a nonabelian 2-cocycle. The correct home for the obstruction is then the *nonabelian second cohomology of the matched pair* $(\mathcal{C}, \mathcal{D}, \lambda)$ — the categorical analogue of the Kac/Masuoka classification of bicrossed (matched-pair) extensions by an H^2 with the two actions entangled [6, 7]. We state this as the open continuation:

Conjecture (nonabelian holonomy). For (L) without (H), the strict factorizations $(\mathcal{C}, \mathcal{D})$ of $\mathcal{K}$ extending $\mathcal{D}$ are classified by a nonabelian H^2 of the Rosebrugh–Wood distributive law λ : the band is the presheaf-of-groups $a \mapsto G_a$ twisted by the left action, the obstruction to a λ is a class in H^2 of that band, and the torsor of λ 's, when nonempty, is acted on simply-transitively by H^1 .

Example 12 shows the abelian theory is genuinely a special case (there (G) happens to hold vacuously, there being no composable $\mathcal{C}$ -pairs; the point is only that $\text{Sk}_{\mathcal{C}}$ already fails to be a category). The decisive positive content of this note is that the *entire* single-category rigid-twist phenomenon — the seed obstruction, and its crown-jewel reading as the laxator — lives in the abelian regime (H), where it is a bona fide H^2 class.

7 Computational verification

All claims are machine-checked by `projects/scratch/cohomology_holonomy.py`, which builds the categories, computes orbits/generators/transversals and the defect (1), assembles the $\mathbb{F}_2$ bar complex (2)–(3), and locates $[\omega]$.

- **Rigid twist.** (L) holds; no transversal closes; $\dim_{\mathbb{F}_2} H^2 = 1$ ($H^2 \cong \mathbb{Z}/2$); all four transversals give a defect outside B^2 representing the *same* nonzero class $((1, 0) \equiv (0, 1) \bmod \langle (1, 1) \rangle)$. Confirms Theorem 10.

- **Pairwise 4-object.** $\mathcal{D}$ non-groupoid; $\dim H^2 = 1$; same nonzero class on all transversals. Confirms Proposition 11.
- **Triple chain** $a \rightarrow x \rightarrow y \rightarrow z$ (**no twist**). (L) and (G) hold; $C^3 \neq \emptyset$ and $\delta^2 \circ \delta^1 = 0$ is verified directly; $\dim H^2 = 0$. Crucially, six of the eight transversals have $\omega \neq 0$ yet $[\omega] = 0$ (they lie in B^2) while a closing transversal *exists* — demonstrating that the obstruction is the class $[\omega]$, not the cochain ω . The full additive 2-cocycle identity is verified on every composable triple.
- **Hypothesis check.** The script verifies abelian vertex groups and the trivial-left-action condition (H)(ii) for the rigid twist and the triple chain (which satisfy (H)(i)) and for the pairwise example (which does not, but is group-valued with trivial left action — Proposition 11). Example 12 is exhibited with $h \circ t \notin \text{orb}(t)$, witnessing that the trivial-left-action condition genuinely restricts.

8 Grant relevance

This upgrades (G) from “a computable obstruction” to “a computable *invariant*.” For a compositional-correctness checker the practical payoff is sharp: after the cheap local freeness test (L), one need not search the whole product of transversal choices for a closed one — one computes a single cohomology class $[\omega] \in H^2(\text{Sk}_{\mathcal{C}}; \mathcal{D})$, and a strict factorization exists iff it vanishes; the dimension of H^2 measures *how* obstructed the system is, and H^1 enumerates the inequivalent factorizations when they exist. In the ga-containers framing this realizes the slogan literally: the diversity-collapse regime $W = 1.0$ is the vanishing of a cohomology class. The honest boundary — that nontrivial target automorphisms push the obstruction into a nonabelian Kac-type H^2 — is the precise next theorem, and ties the program to the bicrossed-product classification literature.

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